

Representation, Reasoning and Logic

Logical Systems Fundamentals

- Formal System Components
 - Syntax (how to make sentences)
 - Semantics (relation to states of affairs)
 - Proof theory (rules for deducing entailments)
- Inference Problem Complexity
 - NP-complete (Cook's Theorem)
 - Polynomial time procedure for Horn sentences
- Improper Logic/Proof Theory results in absurd reasoning

Types of Logic

- Propositional Logic
 - Language: Facts
 - Belief: T / F / Unknown
 - Inference (NP-complete)
- First-Order Logic (FOL)
 - Language: Facts, Objects, Relations
 - Belief: T / F / Unknown
- Other Logics
 - Temporal Logic (Includes Times)
 - Probability Theory (Degree of belief [0..1])
 - Fuzzy Logic (Degree of truth)

Syntax Structure

- Propositional Logic Structure
 - Atom (True, False, AP)
 - Complex Sentence (Connectives: $\wedge, \vee, \Rightarrow, \Leftrightarrow$)
- First-Order Logic Terms
 - Constant (A, 5, Kolkata)
 - Variable (a, x, s)
 - Predicate (Before, HasColor)
 - Function (Mother, Cosine)
- First-Order Logic Sentences
 - AtomicSentence (Predicate, Term ~ Term)
 - Quantifiers ($\forall, \exists$)
 - Connectives ($\Rightarrow, \wedge, \vee, \Leftrightarrow$)

Inference and Reasoning

- Inference Rules (General)
 - Modus Ponens (Implication Elimination)
 - Unit Resolution
 - Resolution ($\alpha \vee \gamma$)
- Inference in FOL
 - Universal Elimination (using ground term substitution)
 - Existential Elimination (new constant not in KB)
 - Existential Introduction
- Logical Reasoning Systems
 - Unification (Makes atomic expressions identical)
 - Resolution (Proving conclusions, requires CNF/Clausal form)
 - Generalized Modus Ponens

Inference Engine Modes

- Forward Chaining (Deduction/Reasoning)
 - Starts with atomic sentences/facts
 - Applies rules forward until goal
 - Properties: Down-up approach, Data-driven
 - Used in: Expert systems (CLIPS)
- Backward Chaining (Deduction/Reasoning)
 - Starts with goal, works backward (Top-down)
 - Uses Modus Ponens
 - Goal is broken into sub-goals
 - Properties: Goal-driven, uses Depth-First Search

Wumpus World Example

- Grid environment, Agent starts bottom-left
- Hazards: Pit, Wumpus (1 each)
- Goal: Gold (1)
- Percepts
 - Stench (In squares containing and adjacent to Wumpus)
 - Breeze (In squares adjacent to Pit)
- Agent Actions/Risks
 - Dies if enters Pit or Wumpus square
 - Can shoot Wumpus along a straight line
 - Has only one arrow